

Erst kommt die KI in
die App, dann die App in
die KI.

ch-open Workshop 2026

ABOUT ME

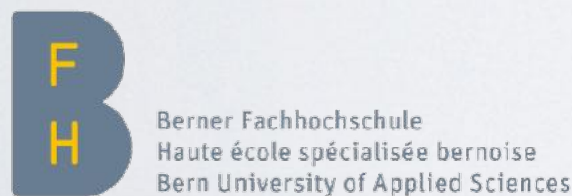
Jonas Bandi

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<https://www.linkedin.com/in/jonasbandi>

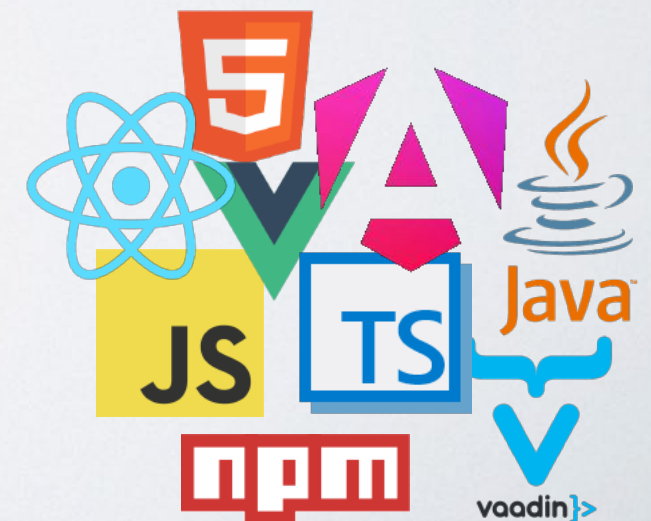


- Freelancer, in den letzten 12 Jahren vor allem in Projekten im Spannungsfeld zwischen modernen Webentwicklung und traditionellen Geschäftsanwendungen.
- Dozent an der Berner Fachhochschule seit 2007
- In-House Kurse & Beratungen zu Web-Technologien im Enterprise: UBS, Postfinance, Mobiliar, AXA, BIT, SBB, Elca, Adnovum, BSI ...



JavaScript / Angular / React / Vue / Vaadin
Schulung / Beratung / Coaching / Reviews

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Disclaimer:

This workshop is vibe coded!



(I am going through the five stages of
AI adoption)

<https://github.com/ivorycode/mcp-chopen-2026/>

I am a fullstack developer with a focus
on modern frontend development.

My expertise is mostly in traditional enterprise systems
(Java, .NET) with modern frontends (React, Angular)

I work on projects, where the end-user
interacts with a browser.

The browser-based user interface is an
abstraction on top of the business functionality.

This talk explores a path into the future where
the last two points are no longer the de-facto
standard ...

The Story:

We used to add chatbots to our apps.
Now we add our apps to the chatbots.

- Kent C. Dodds

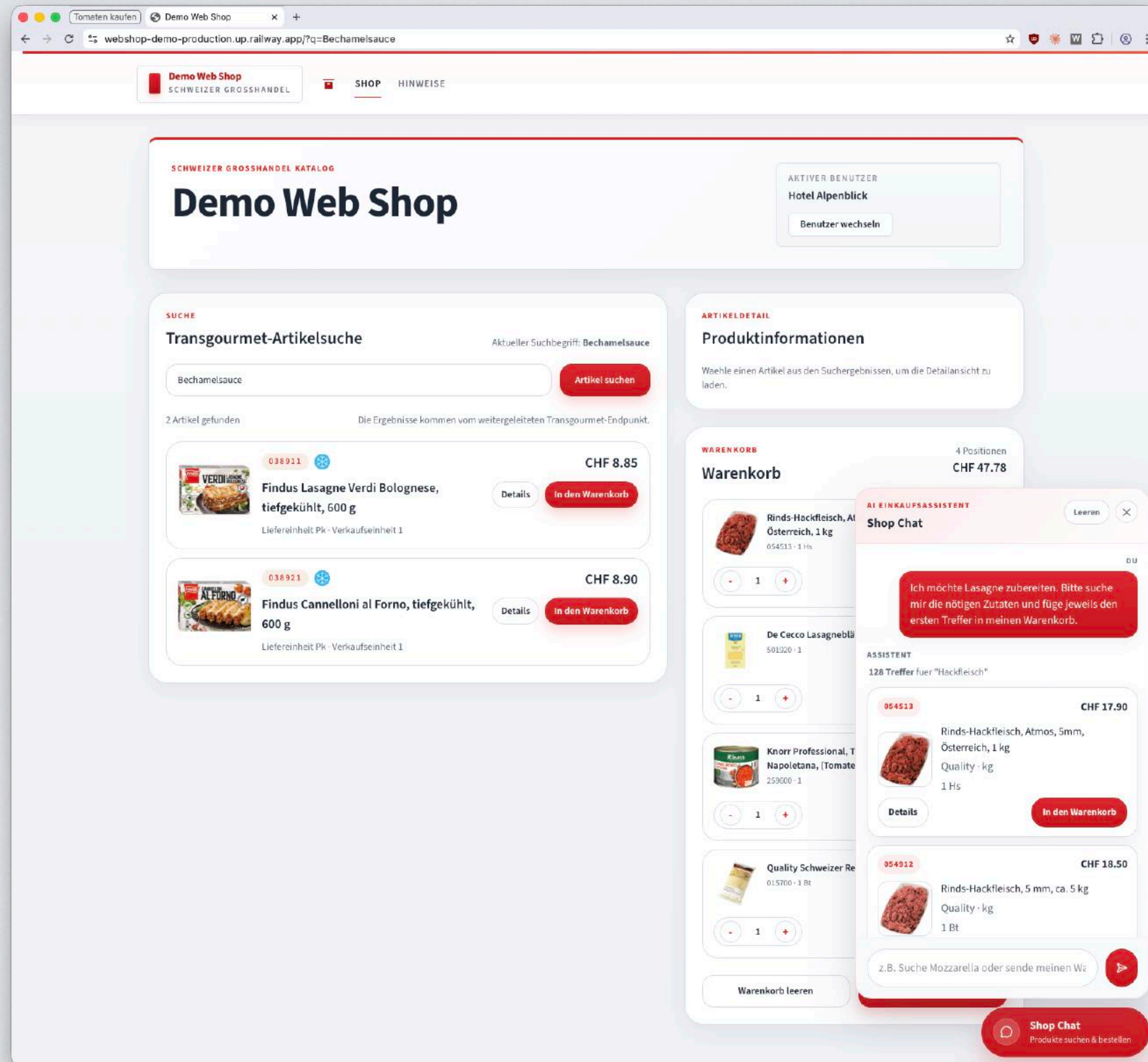
Agenda

Intro - Context & Demo

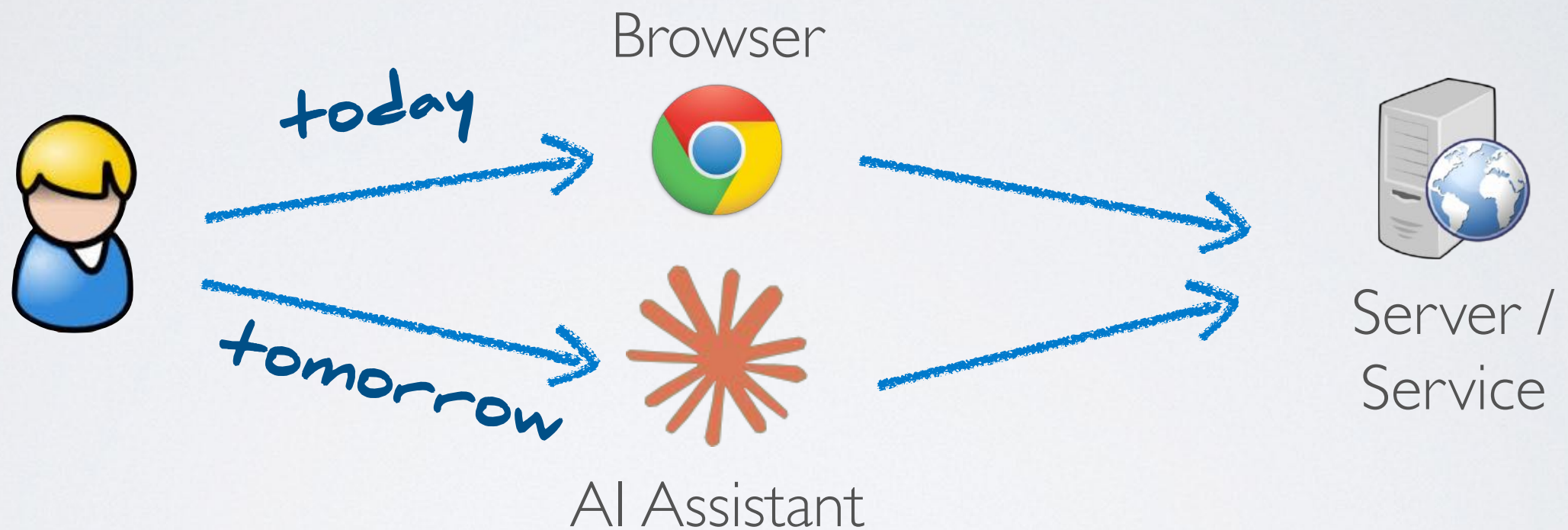
Hands On:

- The AI in the App
- The App in the AI
- WebMCP

Part I: The AI in the App



Thesis:
AI-Agents are the future of user interactions.



Part 2: The App in the AI

Omeletten-Zutaten bei Transgourmet bestellen

Share

Ich möchte Omeletten zubereiten. Bitte suche mir alle Zutaten im Transgourmet-Webshop und füge jeweils den ersten Treffer in meinen Warenkorb.

Searched available tools >

Gerne! Ich suche zuerst alle typischen Omeletten-Zutaten: Eier, Milch, Butter, Mehl, Salz und Pfeffer.

Used Transgourmet Webshop Text Integration >

Alle 5 Zutaten gefunden! Jetzt füge ich jeweils den ersten Treffer in den Warenkorb:

- Eier:** Freshfood Import Bodenhaltung 53g+, 90 Stück (095210)
- Milch:** Quality Vollmilch 3,5% UHT, 12x1l (022600)
- Butter:** Die Butter, 4x250g (027281)
- Mehl:** Weissmehl 10x1kg (228001)
- Salz:** JuraSel Speisesalz grün, 1kg (560053)
- Pfeffer:** Quality Pfeffer schwarz ganz, 400g (512621)

Web-shop-cart-add >

Erledigt! Alle 6 Omeletten-Zutaten wurden erfolgreich in deinen Warenkorb gelegt:

Zutat	Produkt	Preis/Einheit
Eier	Freshfood Import, Bodenhaltung, 53g+, 90 Stück	CHF 0.42/St
Milch	Quality Vollmilch 3,5%, UHT, 12x1l	CHF 1.39/Tp
Butter	Die Butter, 4x250 g	CHF 4.07
Mehl	Weissmehl, 10x1 kg	CHF 1.18/Pk
Salz	JuraSel Speisesalz grün, 1 kg	CHF 1.07
Pfeffer	Quality Pfeffer schwarz ganz, 400 g	CHF 10.51

Ich sehe, dass in deinem Warenkorb noch weitere Artikel von vorherigen Bestellungen sind (Lasagne, Hackfleisch, etc.). Die neuen Omeletten-Zutaten wurden zusätzlich dazu hinzugefügt.

Reply...

Sonnet 4.6

Claude is AI and can make mistakes. Please double-check responses.

AI Agents: More than Chat

Tomaten im Transgourmet-Webshop

chatgpt.com/c/69c0f1a2-f8ac-838f-9d4a-d350cffff4ee

ChatGPT

Memory Off Share

Suche im Transgourmet-Webshop nach Tomaten.

Ich suche direkt im Transgourmet-Webshop nach Tomaten und ziehe die passenden Treffer heraus.

Thought for a few seconds

(s) Called tool

Webshop MCP App

Transgourmet Webshop username guest

Suche Warenkorb

Tomate **Suchen**

284 Ergebnisse für "Tomate"

 040967 QUALITY Cherry Tomaten Rispen, 500 g 1 - Sc - Pack: CHF 3.93	CHF 3.93 MENGE 1 Details 
 040968 QUALITY Cherry Tomaten Rispen, 3 kg 3 - Kt - kg - Pack: CHF 19.98	CHF 6.66 MENGE 1 Details 
 046356 QUALITY Tomaten mit Rispen, ca. 2,5 kg 1 - Sc - kg - Pack: CHF 14.28	CHF 5.71 MENGE 1 Details 

render more

Ich habe die Suche im Transgourmet-Webshop geöffnet. Es gibt 283 Treffer für „Tomaten“.

Ein paar passende Treffer aus den ersten Ergebnissen:

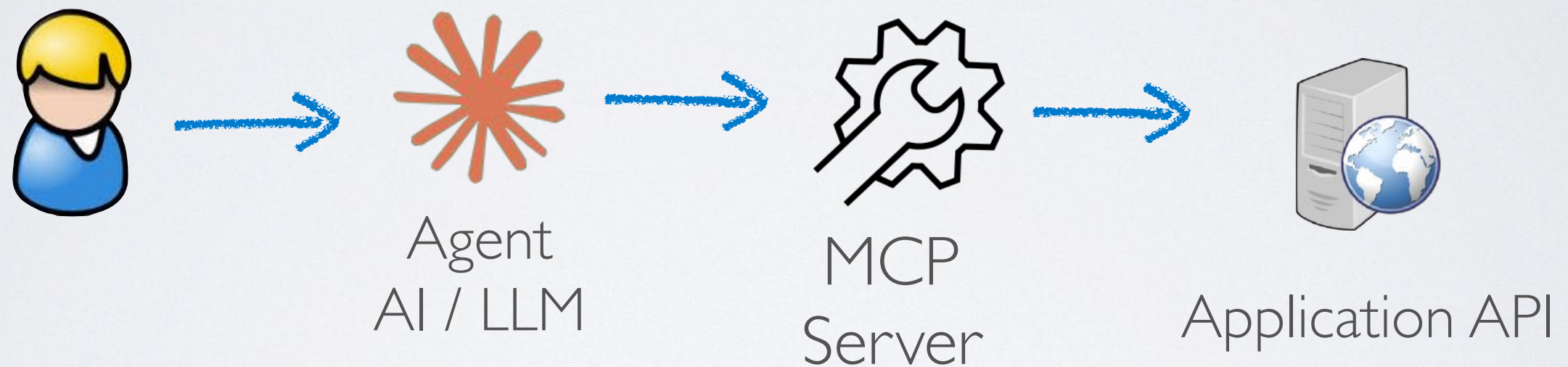
DEVELOPER MODE

Ask anything

+ Thinking

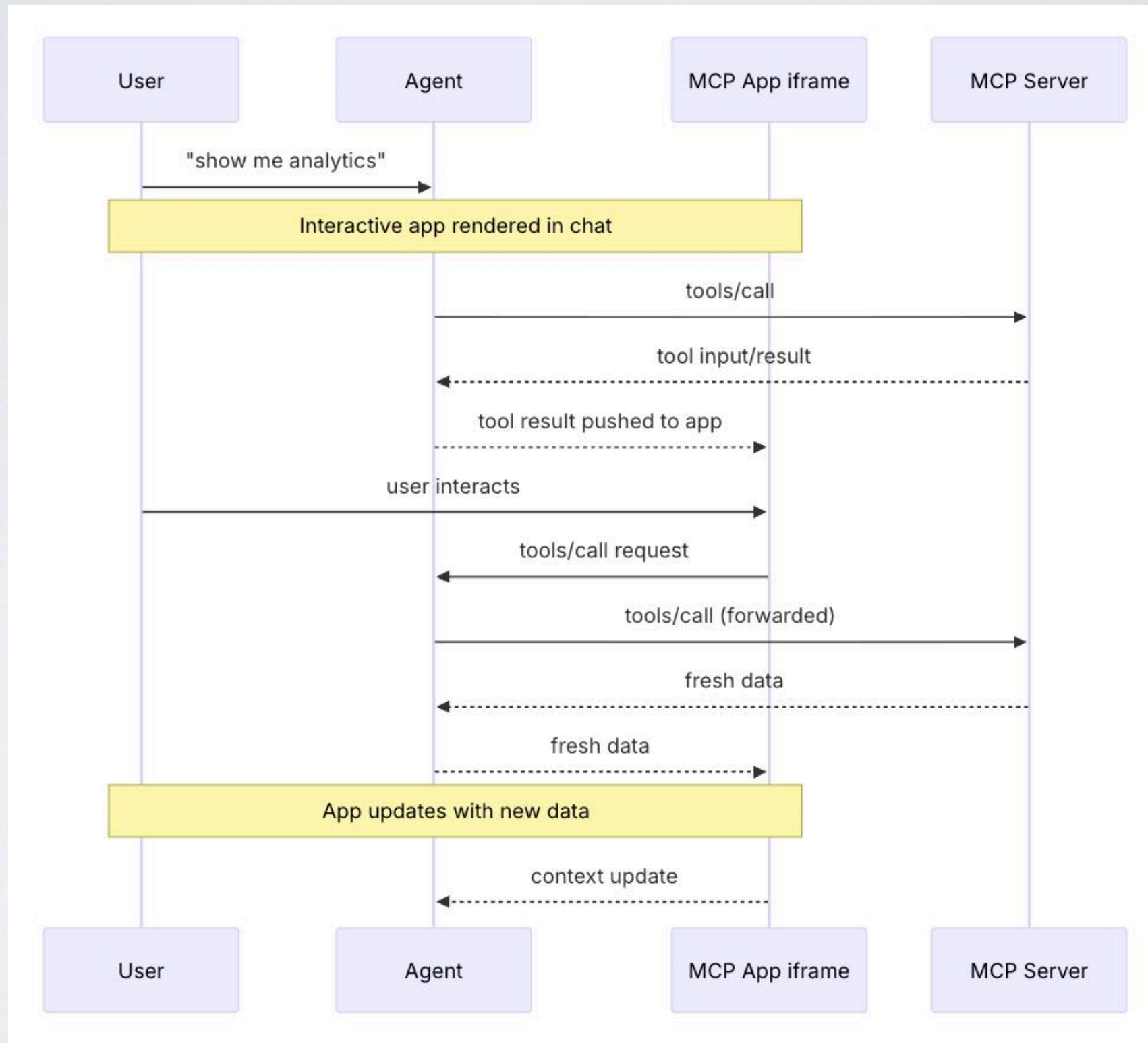
ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#).

Accessing an Application



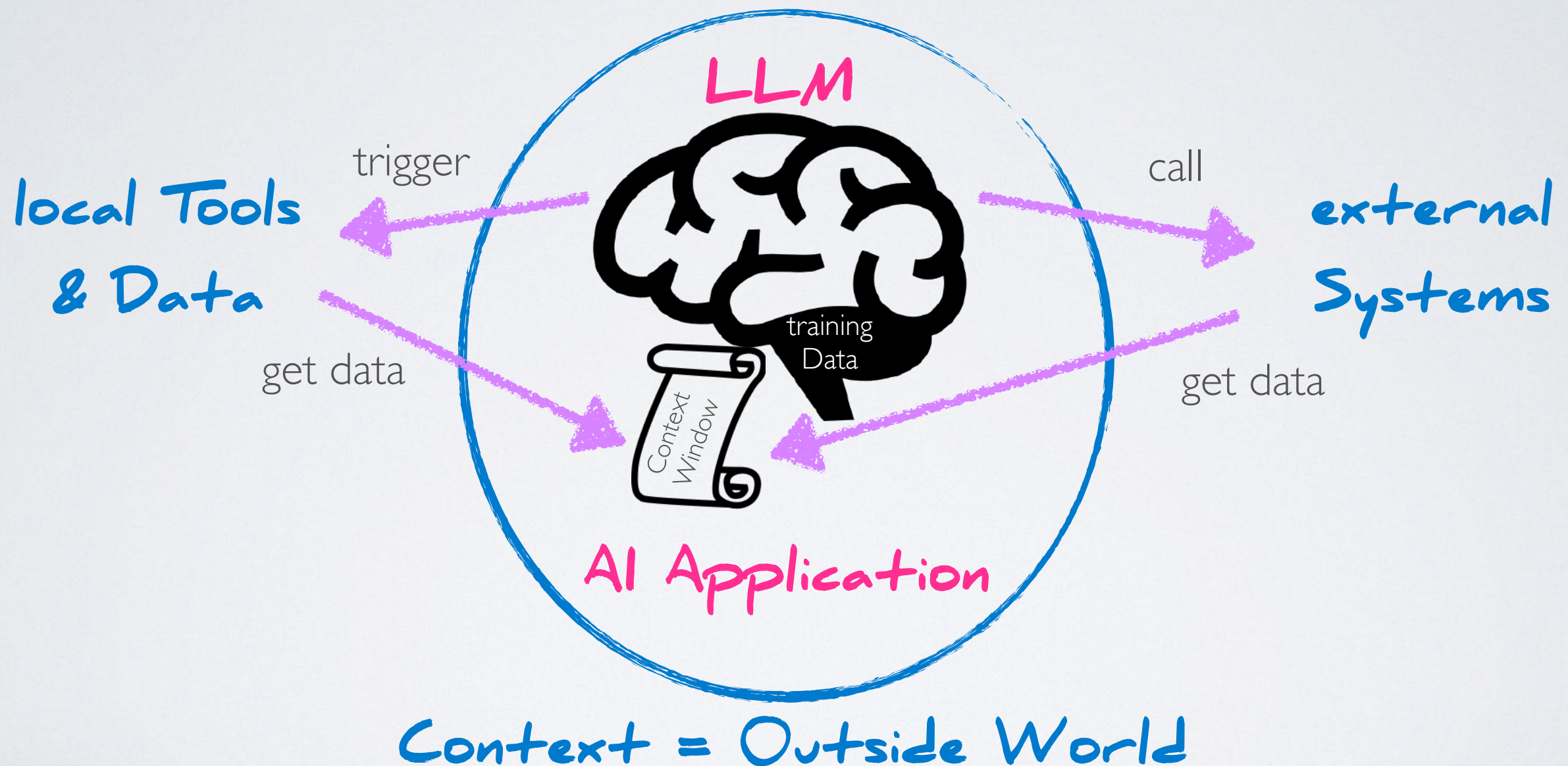


MCP Apps



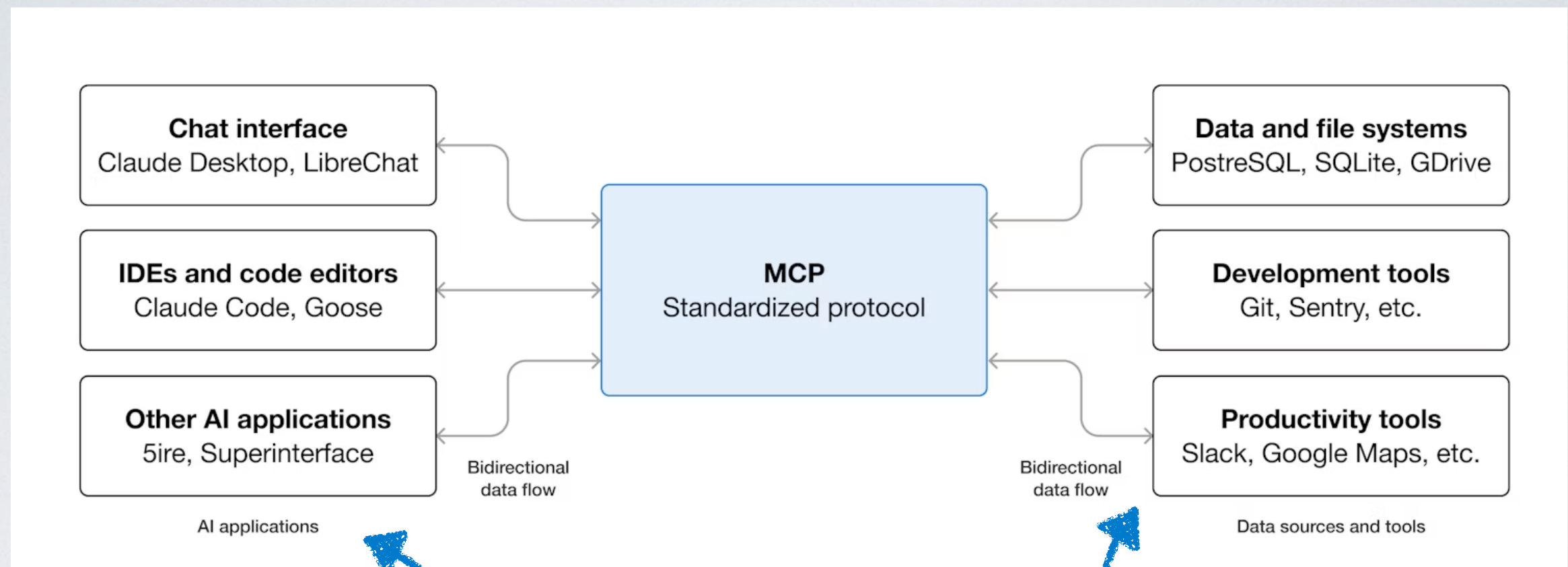
What is MCP?

MCP (Model Context Protocol) is an open-source standard for connecting AI applications to external systems.



MCP standardizes how an LLM interacts with the outside world.

What is MCP?



Somewhere
here is an LLM



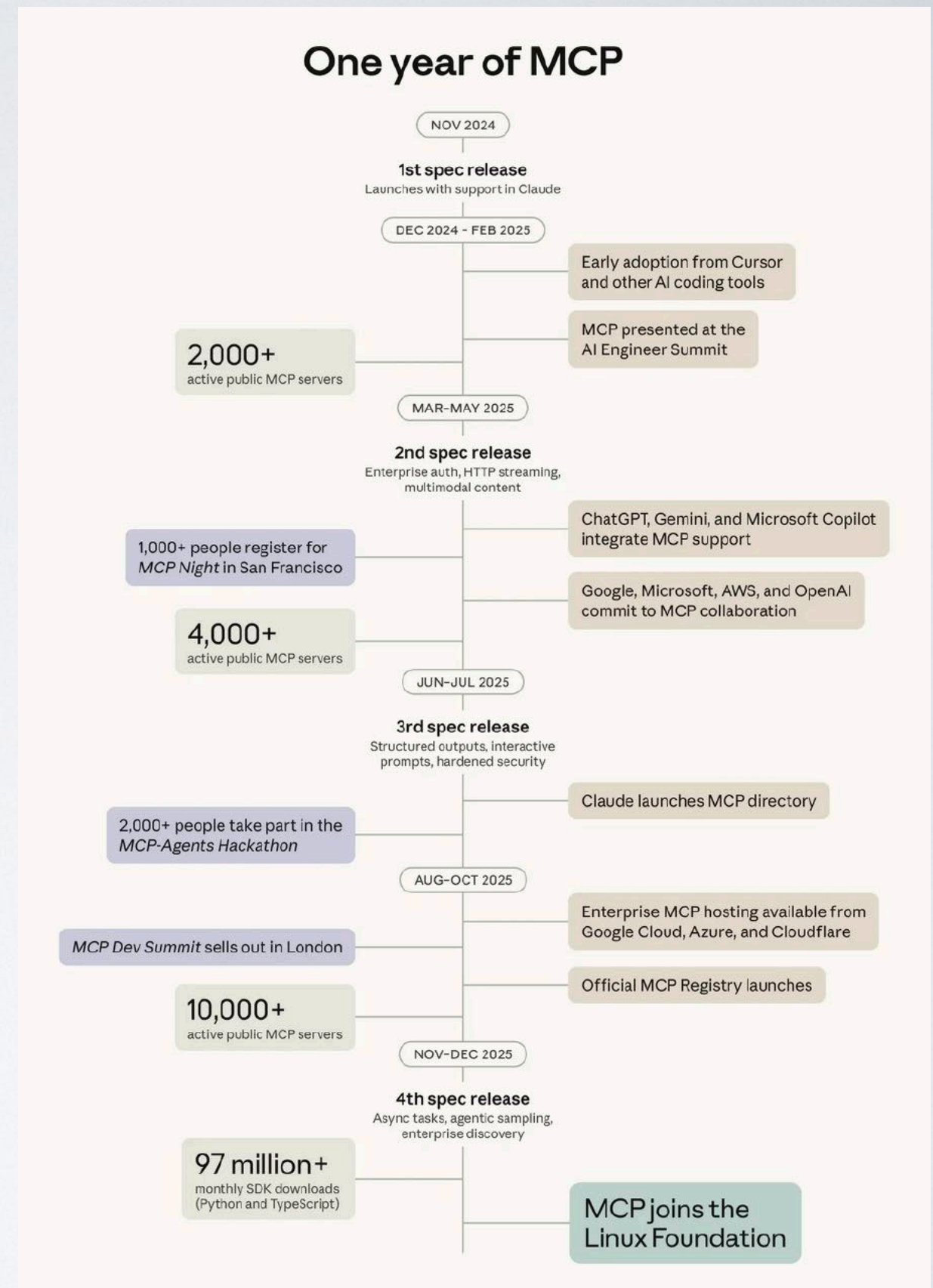
tools exposed by
local or remote
MCP Servers

MCP is a standard/protocol initially created by Anthropic in November 2024

MCP is now part of the Linux Foundation

- In December 2025 the Agentic AI Foundation (AAIF) was founded
- MCP is now developed as an open standard
- 2026-07-28: major spec update

WebMCP is a proposed web standard mainly driven by Google and Microsoft.
It is not part of the MCP standard.





Interactive UI applications that render inside MCP hosts.

Official MCP Extension since January 2026

MCP App evolved out of:

- MCP-UI: <https://mcpui.dev/>
(announced in May 2025)
- OpenAI Apps SDK: <https://developers.openai.com/apps-sdk>
(announced in October 2025)

MCP-UI seems "de-facto" deprecated.

OpenAI App SDK is here to stay.

<https://developers.openai.com/apps-sdk/reference>

<https://developers.openai.com/apps-sdk/mcp-apps-in-chatgpt>

<https://github.com/modelcontextprotocol/modelcontextprotocol/pull/1865>

AND NOW FOR
SOMETHING
COMPLETELY
DIFFERENT



WebMCP

Web MCP

A proposed Web Standard / Early Preview

Announced on February 9, 2026

https://docs.google.com/document/d/1rtUIfRPS0bMqd9abMG_hc6K9OAI6soUy3Kh00toAgyk/

A way for web pages to expose structured tools for AI-agents.
Improving the reliability and performance of interactions
between web page and AI-agent.

Similarity to MCP:

- It is a mechanism to expose tools to an agent

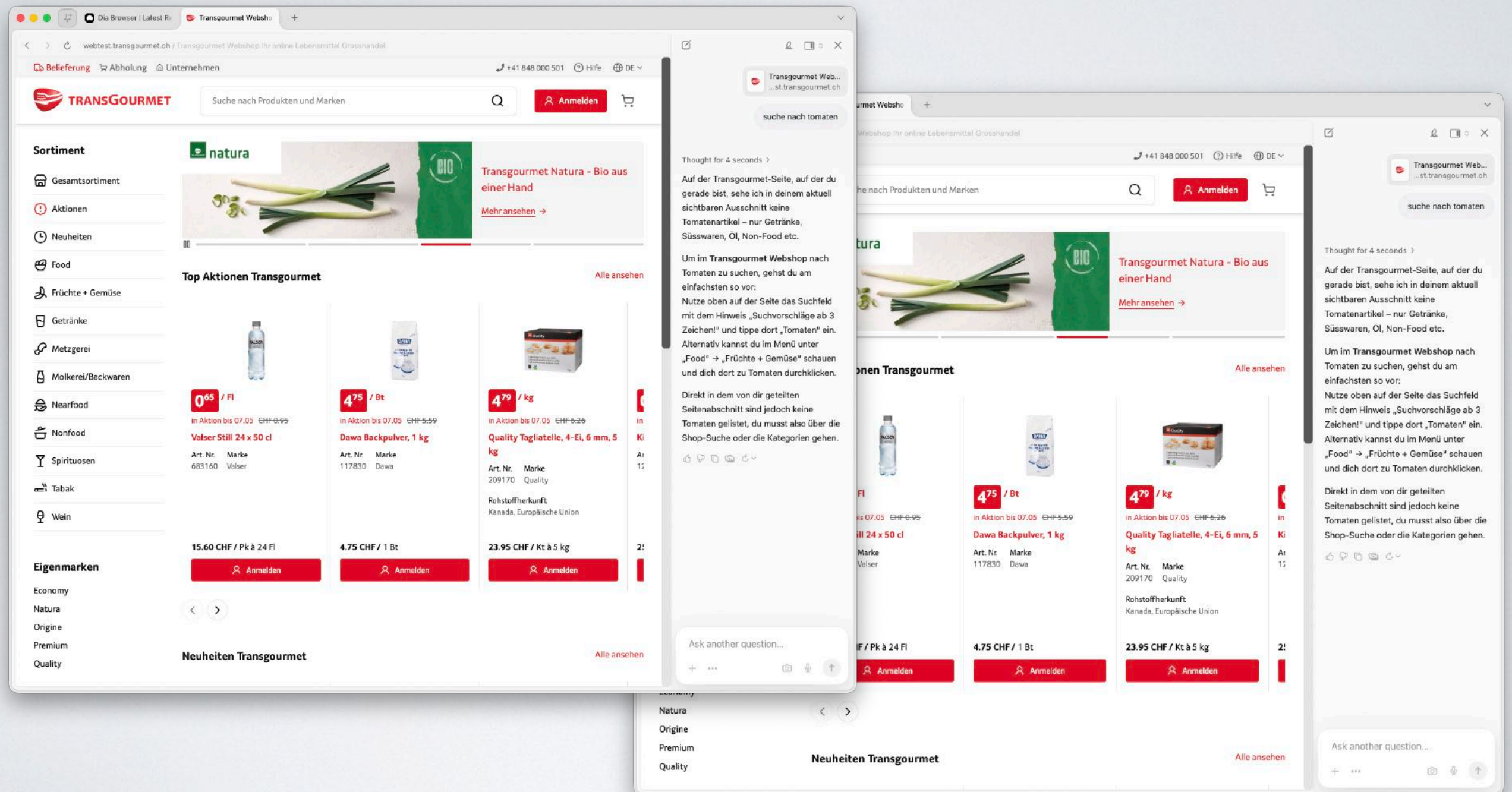
Differences:

- client-side mechanism
- DOM-API
- no "transport protocol" involved

Changelog

DATE	CHANGES
Apr 24, 20...	- Added readOnlyHint and untrustedContentHint annotations (source)
Mar 27, 20...	- Added registerTool() method signal option and removed unregisterTool() (source) - Updated input groups (radio, etc) toolparamdescription using nearest <fieldset> (source) - Switched from oneOf to anyOf in the computed input schema example. (source)
Mar 12, 20...	- Removed toolparamtitle attribute - Added demo
Mar 6, 2026	- Removed provideContext and clearContext methods
Feb 10, 20...	- Declarative API tweaks - Added Demo resources section
Feb 9, 2026	- Initial version

"Smart" Browsers



Dia, Atlas, Comet ... Claude Browser Extension ...

Chapter 3: Agent controlled App

The screenshot displays a web browser window with the address bar showing `webshop-demo-production.up.railway.app/?q=Milch`. The main content area shows the 'Demo Web Shop' interface, which includes a search bar with the query 'Milch', a list of search results for various milk products, and a shopping cart with 4 items totaling CHF 47.78. The products listed include 'Quality Vollmilch 3,5%, UHT, 12 x 1 l' (CHF 1.39), 'Economy Vollmilch 3,5%, UHT, 8 x 1,5 l' (CHF 1.89), 'Emmi good day Milch laktosefrei 1,5%, UHT, 12 x 1 l' (CHF 2.16), 'Emmi Vollmilch, 3,5%, UHT, 12 x 1 l' (CHF 1.57), 'Lovely Vollmilch 3,5%, UHT, 6 x 1,5 l' (CHF 2.43), 'Economy Milchdrink 1,5%, UHT, 8 x 1,5 l' (CHF 1.74), and 'Lovely Vollmilch 3,5%, UHT, Stelo, 2 x 6' (CHF 1.61).

Overlaid on the right side of the browser window is the 'WebMCP - Model Context Tool Inspector' extension. It features a table of 'WebMCP Tools' with columns for 'description', 'inputSchema', and 'name'. The tools listed are:

description	inputSchema	name
Fuege einen Artikel ueber seine Artikelnummer in den Warenkorb ein.	<pre>{ "type": "object", "properties": { "articleNumber": { "type": "string", "description": "Exact product article number to add to the cart. ", "quantity": { "type": "number", "description": "Quantity to add. Defaults to 1." } }, "required": ["articleNumber"] }</pre>	addArticleToCart
Lade den aktuellen Warenkorb des angemeldeten Demo-Benutzers.	<pre>{ "type": "object", "properties": { } }</pre>	getCart
Suche nach passenden Produkten im Shop-Katalog fuer einen Suchbegriff.	<pre>{ "type": "object", "properties": { "term": { "type": "string", "description": "Search term for products in the webshop catalog." } }, "required": ["term"] }</pre>	searchProducts
Sende den aktuellen Warenkorb ueber den bestehenden Checkout-Prozess.	<pre>{ "type": "object", "properties": { } }</pre>	sendCart

Below the table, there are buttons for 'Copy as ScriptToolConfig' and 'Copy as JSON'. The 'Interact with the Page' section contains a 'User Prompt' field with the text 'Suche die Zutaten für Lasagne und lege den jeweils günstigsten Artikel in den Warenkorb.' and buttons for 'Send', 'Reset', 'Set Gemini API key', and 'Copy trace'. The 'Tool' dropdown is set to 'addArticleToCart', and the 'Input Arguments' field contains `{ "articleNumber": "example_string", "quantity": 0 }`. An 'Execute Tool' button is located at the bottom of the tool inspector.

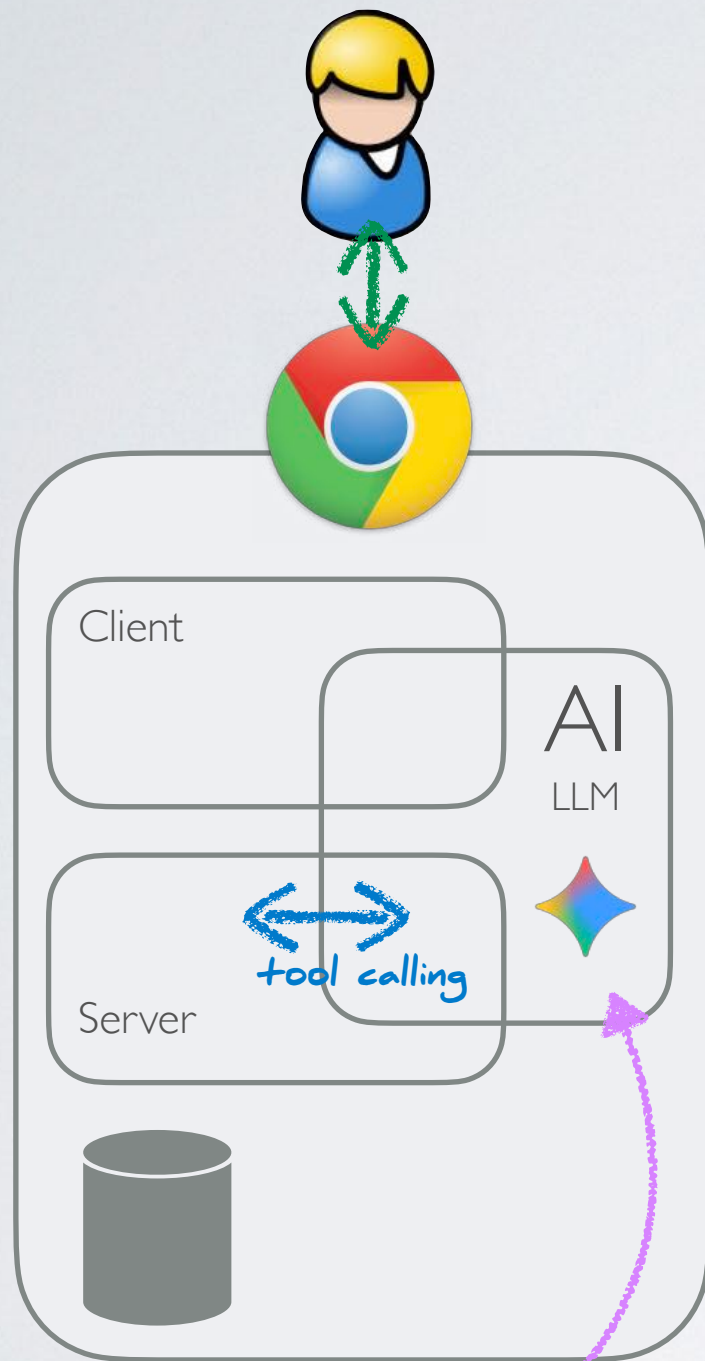
```
navigator.modelContextTesting.executeTool("searchProducts", JSON.stringify({term: 'Milch'}))
```


Summary

Three Mechanisms how AI can control our app:

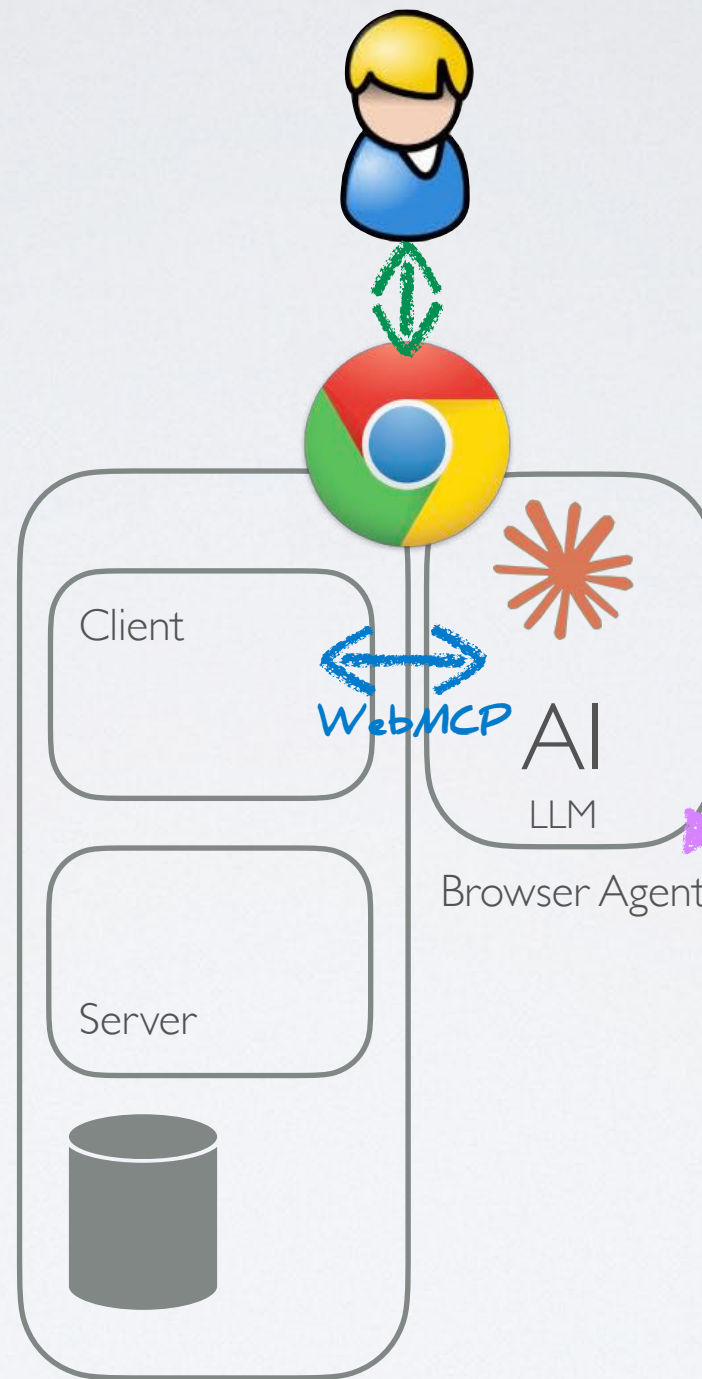
- AI inside the App
- The App in the Agent
- The agent-controlled App

AI in the App



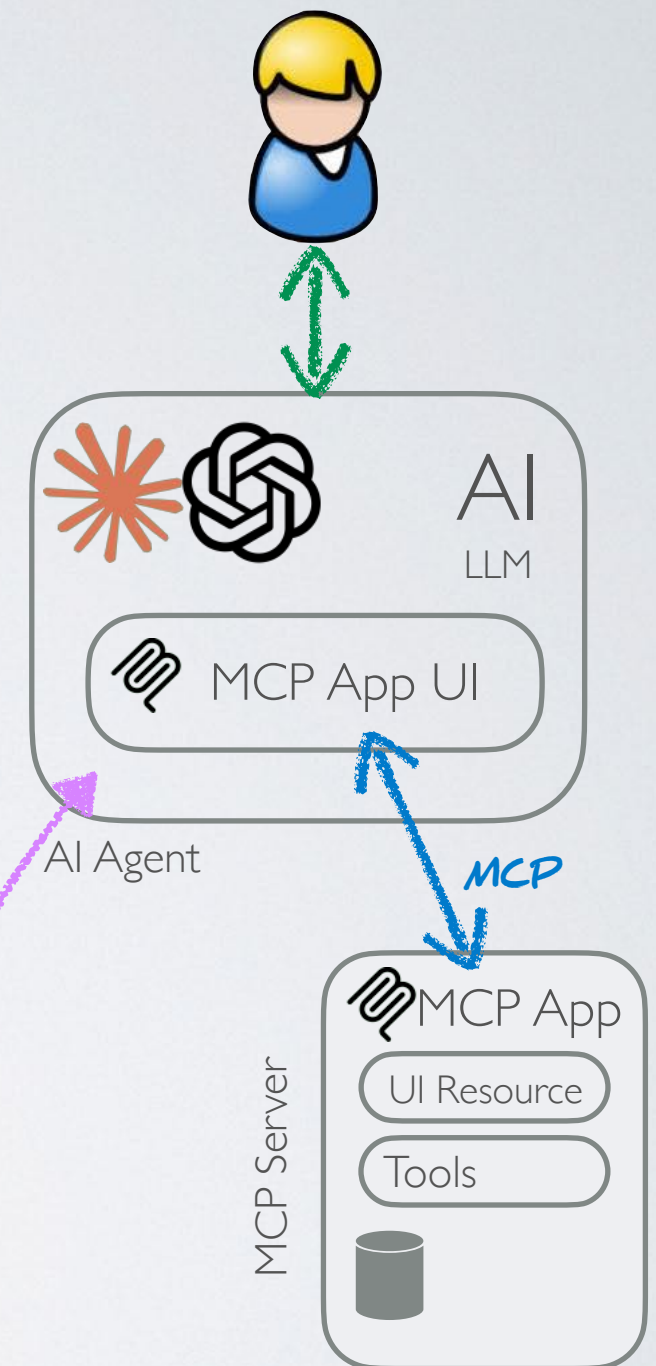
App pays for LLM

AI-controlled App



User pays for LLM

MCP App



New UI Paradigms?

How do you "navigate" in a chat-bot?

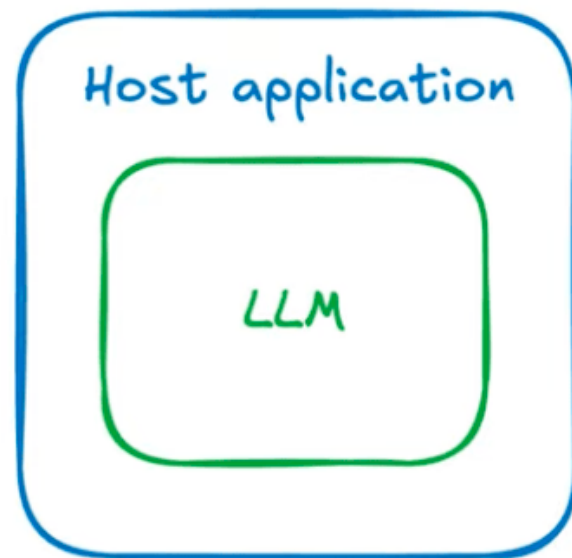
"Components will kill Pages ..."

AI will be the new steering wheel of the internet, it won't be navigating pages.

History of Tool Calling

Phase 1

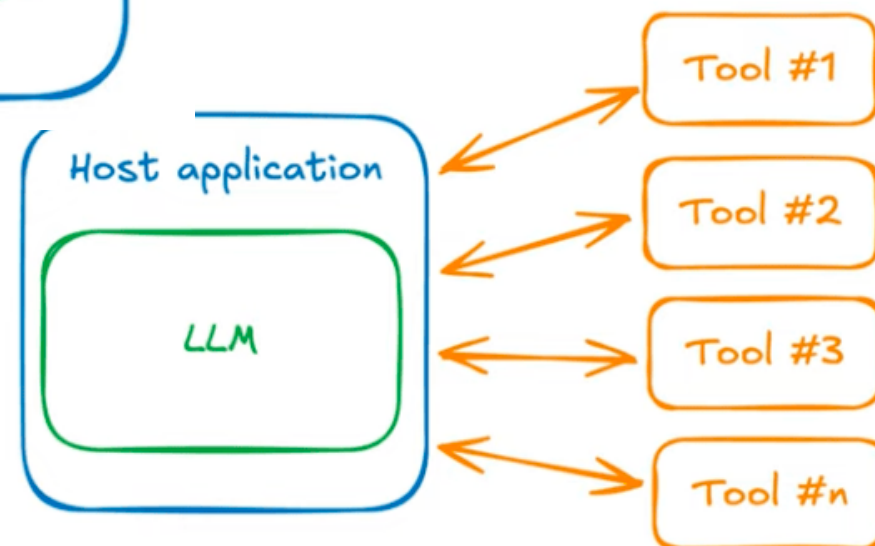
Answers questions 🤖



2022

Phase 2

Can do stuff 🤖

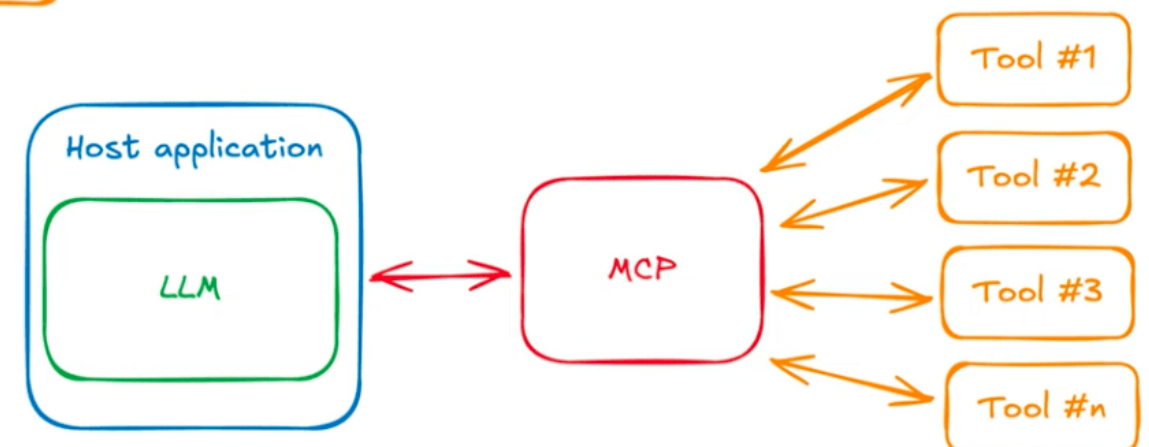


2023

Models are trained to call tools.
Often optimized for a specific harness.
Same model + different harness = vastly different results.

Phase 3

Can do ANYTHING 🤖



2024

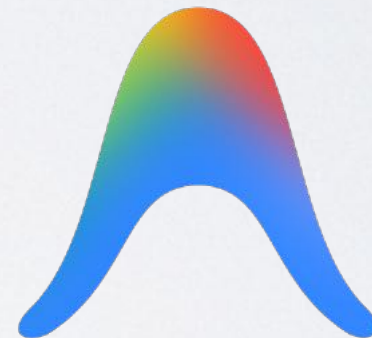
MCP is the "USB port for AI"

Have fun with AI!

Slides & Code:

<https://github.com/jbandi/agentik-sbb-2026>

<https://github.com/jbandi/agentik-sbb-example>



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